Sequences and Basic Series Convergence Tests

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the interval of convergence of $\sum_{n=0}^{\infty} x^n$.

Theorem (Interval of Convergence for Geometric Power Series): The series $\sum_{n=0}^{\infty} x^n$ is a geometric series with common ratio $r = x$. Step 1 · Apply Geometric Convergence Criterion: An infinite geometric series converges if and only if $|r| < 1$. Therefore: $|x| < 1 \Rightarrow -1 < x < 1$. Radius of convergence is $R = 1$, centered at $c = 0$. Step 2 · Test the boundary endpoints: At $x = 1$: $\sum 1^n = 1 + 1 + 1 + \ldots$ diverges by the nth-term test ($\lim_{n\to\infty} 1 = 1e0$). At.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{2^n}$.

Theorem (Ratio Test for Radius of Convergence): Step 1 · Set up ratio limit for $a_n = \frac{x^n}{2^n}$: $L = \lim_{n\to\infty} \left|\frac{x^{n+1}}{2^{n+1}} \cdot \frac{2^n}{x^n}\right| = \lim_{n\to\infty} \left|\frac{x \cdot 2^n}{2 \cdot 2^n}\right| = \frac{|x|}{2}$. Step 2 · Apply convergence condition $L < 1$: $\frac{|x|}{2} < 1 \Rightarrow |x| < 2$. Conclusion: The series converges for all x strictly within distance 2 of the center $c = 0$. The radius of convergence is: $\mathbf{R = 2}$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Does $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converge?

Theorem (Alternating Series Test / Leibniz's Theorem): An alternating series of the form $\sum_{n=1}^{\infty} (-1)^n b_n$ or $\sum_{n=1}^{\infty} (-1)^{n+1} b_n$ converges if all three conditions are satisfied: 1. $b_n > 0$ for all n; 2. $b_{n+1} \leq b_n$ for all $n \geq N$ (sequence of magnitudes is monotonically decreasing); 3. $\lim_{n\to\infty} b_n = 0$. Verification for $b_n = \frac{1}{n}$: 1. $b_n = \frac{1}{n} > 0$ for all integers $n \geq 1$. 2. $b_{n+1} = \frac{1}{n+1} < \frac{1}{n} = b_n$ holds.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Find the speed of a particle with $x = t^2,\ y = t^2$ at $t = 1$.

Theorem (Particle Kinematics & Speed Formula): For a particle whose position vector is $\vec{r}(t) = \langle x(t), y(t)\rangle$, the velocity vector is $\vec{v}(t) = \langle x'(t), y'(t)\rangle$, and its instantaneous speed is the Euclidean magnitude of velocity: $\text{Speed} = ||\vec{v}(t)|| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$. Step 1 · Determine velocity components: $x'(t) = \frac{d}{dt}[t^2] = 2t, \quad y'(t) = \frac{d}{dt}[t^2] = 2t$. Step 2 · Evaluate components at.

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5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Does $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converge or diverge?

Theorem (p-Series Convergence Test): The infinite series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$, and diverges if $p \leq 1$. Step 1 · Identify the series parameter p: The general term is $a_n = \frac{1}{n^2}$, which is a p-series with exponent $p = 2$. Step 2 · Test against convergence condition: Because $p = 2 > 1$, the series converges by the p-Series Test. Analytical Note: By Euler's classical Basel solution, this series converges to the exact sum.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Unit 10 · Infinite Sequences & Series — Convergence Tests · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 3 · Advanced / AP FRQOn an AP Calculus BC exam question asking whether a series $\sum_{n=1}^{\infty} a_n$ converges, a student writes: "Since $\lim_{n\to\infty} a_n = 0$, the series converges." Evaluate the mathematical validity of this statement, identify the logical fallacy, and provide a standard counterexample that disproves it.

Evaluation: The statement is mathematically invalid and earns 0 points. Logical Fallacy: The student has committed the fallacy of the converse. The n-th Term Test states: If a series $\sum a_n$ converges, then $\lim_{n\to\infty} a_n = 0$. The contrapositive is also true: If $\lim_{n\to\infty} a_n eq 0$, then the series diverges. However, $\lim_{n\to\infty} a_n = 0$ is a necessary condition, NOT a sufficient condition for convergence. Counterexample: The harmonic series.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · FoundationsFind the radius of convergence R for the power series $\sum_{n=0}^{\infty} n!\,(2x-1)^n$.

Apply the Ratio Test: $L = \lim_{n\to\infty} \left| \frac{(n+1)!\,(2x-1)^{n+1}}{n!\,(2x-1)^n} \right| = |2x-1| \lim_{n\to\infty} (n+1)$. For any $x eq \frac{1}{2}$, this limit is ∞. The series converges only at its center $x = \frac{1}{2}$. Thus, the radius of convergence is $\mathbf{R = 0}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 2 · MediumThe graph of f' consists of a semicircle of radius 2 centered at the origin, lying above the x-axis. If $f(0) = 5$, find the value of $f(2)$.

By the Fundamental Theorem of Calculus: $f(2) = f(0) + \int_0^2 f'(x)\,dx$. The region under f' from $x = 0$ to $x = 2$ is a quarter-circle of radius 2: $\int_0^2 f'(x)\,dx = \frac{1}{4}\pi(2^2) = \pi$. Therefore, $f(2) = 5 + \pi \approx \mathbf{8.142}$.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 1 · FoundationsOn a Rate In / Rate Out FRQ, if rain falls into a cistern at rate $R(t)$ and is pumped out at rate $P(t)$, when is the amount of water in the cistern at a relative maximum?

Let $W(t)$ be the amount of water. $W'(t) = R(t) - P(t)$. A relative maximum occurs when $W'(t) = 0$ and changes from positive to negative, which occurs when $R(t) = P(t)$ and $R(t)$ crosses below $P(t)$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 1 · FoundationsHow many total questions are on the AP Calculus AB exam, and what percentage of the composite score comes from Free-Response questions?

The exam consists of 45 Multiple-Choice Questions and 6 Free-Response Questions (51 total). Free Response accounts for exactly 50% of the composite score (54 out of 108 scaled points).

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[AP Calculus AB Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 2 · MediumFind the third-degree Taylor polynomial $P_3(x)$ for $f(x) = \sqrt{x}$ centered at $c = 4$.

Evaluate derivatives at $c = 4$: • $f(x) = x^{1/2} \Rightarrow f(4) = 2$. • $f'(x) = \frac{1}{2}x^{-1/2} \Rightarrow f'(4) = \frac{1}{2(2)} = \frac{1}{4}$. • $f''(x) = -\frac{1}{4}x^{-3/2} \Rightarrow f''(4) = -\frac{1}{4(8)} = -\frac{1}{32}$. • $f'''(x) = \frac{3}{8}x^{-5/2} \Rightarrow f'''(4) = \frac{3}{8(32)} = \frac{3}{256}$. Assemble the Taylor polynomial: $P_3(x) = 2 + \frac{1}{4}(x-4) + \frac{-1/32}{2!}(x-4)^2 + \frac{3/256}{3!}(x-4)^3$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 3 · Advanced / AP FRQLet $P_4(x)$ be the fourth-degree Maclaurin polynomial for $\cos(x)$. Use the Lagrange Error Bound to prove that \(|\cos(0.5) - P_4(0.5)|

For $f(x) = \cos(x)$, $f^{(5)}(x) = -\sin(x)$. Since $|-\sin(t)| \le 1$ for all t, we take $M = 1$. Apply the Lagrange Error Bound formula with $n = 4, c = 0, x = 0.5$: $|R_4(0.5)| \le \frac{M}{(4+1)!}|0.5 - 0|^{4+1} = \frac{1}{5!}(0.5)^5 = \frac{1}{120}\left(\frac{1}{32}\right) = \frac{1}{3840}$. Because $3840 > 3000$, it follows that: \[\frac{1}{3840}

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals**

[Unit 10 · Power Series, Taylor & Maclaurin Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Euler's Method and Logistic Growth**

[AP Calculus BC Exam Format & Units Overview · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Parametric Equations**

Level 3 · Advanced Find the full interval of convergence (including endpoint behavior) for the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n^2 \cdot 4^n}$.

Apply the Ratio Test: $L = \lim_{n\to\infty} \left|\frac{(2x-1)^{n+1}}{(n+1)^2 4^{n+1}} \cdot \frac{n^2 4^n}{(2x-1)^n}\right| = \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^2 \cdot \frac{|2x-1|}{4} = \frac{|2x-1|}{4}$ Require x: $2x - 1 = 4$, giving $\sum_{n=1}^{\infty} \frac{4^n}{n^2 4^n} = \sum_{n=1}^{\infty} \frac{1}{n^2}$, which converges by the p-series test ($p = 2 > 1$). Endpoint $x = -3/2$: $2x - 1 = -4$, giving.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.02 Vector-Valued Functions**

Level 2 · Medium Determine whether the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges absolutely, converges conditionally, or diverges.

Step 1: Test for absolute convergence. Consider $\sum |a_n| = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$. This is a p-series with $p = 1/2 \le 1$, which diverges. Hence the series does NOT converge absolutely. Step 2: Apply Alternating Series Test. Here $b_n = \frac{1}{\sqrt{n}} > 0$. 1. Monotonicity: \(n+1 > n \implies \sqrt{n+1} > \sqrt{n} \implies \frac{1}{\sqrt{n+1}} conditionally convergent.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Polar Curves and Derivatives**

Level 1 · Foundations Use the Divergence Test to determine whether $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$ converges or diverges.

Compute the limit of the general term as $n \to \infty$: $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \cos\left(\frac{1}{n}\right) = \cos\left(\lim_{n\to\infty} \frac{1}{n}\right) = \cos(0) = 1$ Because $\lim_{n\to\infty} a_n = 1e0$, the series diverges by the Divergence Test.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.02 Area in Polar Coordinates**

[Convergence Tests & Power Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Determine the limit of the sequence $a_n = \frac{4n^2 - 3n}{2n^2 + 5}$.

a. 2 ✓

b. 4

c. ∞ (diverges)

d. 0

Divide numerator and denominator by n^2: $\lim_{n\to\infty} \frac{4 - 3/n}{2 + 5/n^2} = \frac{4}{2} = 2$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which statement is guaranteed by the Monotone Convergence Theorem for Sequences?

a. **Every bounded monotonic sequence converges** ✓

b. Every alternating sequence converges

c. Every bounded sequence is monotonic

d. Every convergent sequence is positive

The Monotone Convergence Theorem states that if a sequence is bounded and monotonic (either non-decreasing or non-increasing), it must converge.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Evaluate the sum of the series $\sum_{n=1}^{\infty} 3(\frac{2}{5})^n$.

a. 2 ✓

b. $\frac{6}{5}$

c. $\frac{15}{2}$

d. $\frac{5}{2}$

First term $a = 3(\frac{2}{5}) = \frac{6}{5}$, common ratio $r = \frac{2}{5}$. Sum $= \frac{a}{1-r} = \frac{6/5}{1 - 2/5} = \frac{6/5}{3/5} = 2$.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $\lim_{n\to\infty} a_n = \frac{1}{2}$, what does the n-th Term Test conclude about $\sum_{n=1}^{\infty} a_n$?

a. The series diverges ✓

b. The series converges to $\frac{1}{2}$

c. The test is inconclusive

d. The series is geometric

The n-th Term Divergence Test states that if $\lim_{n\to\infty} a_n \neq 0$, the series must diverge.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $\lim_{n\to\infty} a_n = 0$, what can be concluded about $\sum a_n$?

a. The test is inconclusive; the series may converge or diverge ✓

b. The series converges

c. The series diverges

d. The sum is zero

Having terms approach zero is a necessary condition for convergence, but not sufficient (e.g., the harmonic series $\sum \frac{1}{n}$ has terms approaching 0 but diverges).

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what values of p does the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?

a. $p > 1$ **✓**

b. $p \geq 1$

c. $p < 1$

d. $p > 0$

By the p-series test, $\sum \frac{1}{n^p}$ converges if and only if $p > 1$. For $p \leq 1$, it diverges.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which series converges by the Integral Test?

a. $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ ✓

b. $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

c. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

d. $\sum_{n=1}^{\infty} \frac{1}{2n+1}$

$\int_2^{\infty} \frac{1}{x(\ln x)^2} dx = [-\frac{1}{\ln x}]_2^{\infty} = \frac{1}{\ln 2} < \infty$. Since the improper integral converges, the series converges.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What three conditions must $f(x)$ satisfy on $[1, \infty)$ to apply the Integral Test to $\sum a_n$?

a. $f(x)$ **must be continuous, positive, and decreasing** ✓

b. $f(x)$ must be differentiable, alternating, and bounded

c. $f(x)$ must be positive and increasing

d. $f(x)$ must be a rational function

The hypotheses of the Integral Test require $f(x)$ to be continuous, positive, and decreasing on $[1, \infty)$ where $f(n) = a_n$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^3+5n}$.

a. Converges by Direct Comparison with $\sum \frac{1}{n^3}$ ✓

b. Diverges by Comparison with $\sum \frac{1}{n}$

c. Diverges by n-th term test

d. Inconclusive

For all $n \geq 1$: $\frac{1}{n^3+5n} < \frac{1}{n^3}$. Since $\sum \frac{1}{n^3}$ is a convergent p-series ($p = 3 > 1$), the given series converges by Direct Comparison.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which series should be used for the Limit Comparison Test with $\sum_{n=1}^{\infty} \frac{3n+2}{2n^3-1}$?

a. $\sum \frac{1}{n^2}$ (convergent) ✓

b. $\sum \frac{1}{n}$ (divergent)

c. $\sum \frac{1}{n^3}$ (convergent)

d. $\sum n$ (divergent)

The dominant terms are $\frac{3n}{2n^3} = \frac{3}{2n^2}$. Comparing with $b_n = \frac{1}{n^2}$ yields a positive finite limit $\frac{3}{2}$, confirming convergence.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the Limit Comparison Test, if $\lim_{n\to\infty} \frac{a_n}{b_n} = L$ where $0 < L < \infty$, what does this mean?

a. Both $\sum a_n$ and $\sum b_n$ either both converge or both diverge ✓

b. $\sum a_n$ converges to L

c. $\sum a_n$ converges while $\sum b_n$ diverges

d. The test is inconclusive

A finite positive limit L ensures that both series share the exact same convergence or divergence behavior.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The alternating series $\sum(-1)^n\frac{1}{n}$:

a. Diverges

b. Converges absolutely

c. **Converges conditionally ✓**

d. Oscillates without limit

Alternating harmonic series converges conditionally.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x)$ whose graph is shown below. The graph consists of a line segment and a quarter circle. The line segment connects the points $(2, 4)$ and $(4, 0)$. The quarter circle is in the second quadrant and is centered at $(0, 0)$ with radius 2. Let $A(x)$ be the area of the region bounded by the graph of $f(t)$, the t-axis, and the lines $t = 0$ and $t = x$. (a) Find $A(3)$.

a. **4 ✓**

b. 6

c. 5

d. 7

The function $f(x)$ is defined piecewise. For $0 \le t \le 2$, $f(t)$ is the upper part of a circle centered at the origin with radius 2, so $f(t) = \sqrt{4 - t^2}$. For $2 < t \le 4$, $f(t)$ is a line segment connecting $(2, 4)$ and $(4, 0)$. The equation of this line can be found using the two-point form: $y - 0 = \frac{4-0}{2-4}(x - 4)$, which simplifies to $y = -2(x - 4)$, so $f(t) = -2(t - 4)$. To find $A(3)$, we need to calculate the definite integral $\int_0^3 f(t)\,dt$. We split this integral i.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of a differentiable function f is shown in the figure above. The function f has a local maximum at $x = c$ and a local minimum at $x = e$. The point $(c, f(c))$ is approximately $(1, 4)$ and the point $(e, f(e))$ is approximately $(4, -2)$. The shaded region, bounded by the graph of f, the x-axis, and the lines $x = 0$ and $x = 5$, has area 12. $If\int_{0}^{c} f(x) \, dx = k, whatisthevalueof\int_{c}^{5} f(x) \, dxintermsofk$?

a. 12 + k

b. **12 - k ✓**

c. k - 12

d. k + 2

The total area of the shaded region from $x = 0$ to $x = 5$ is given as 12. This total area can be expressed as the sum of the areas of two sub-regions: the area from $x = 0$ to $x = c$ and the area from $x = c$ to $x = 5$. Mathematically, this can be written using definite integrals: $\int_0^5 f(x)\,dx = \int_0^c f(x)\,dx + \int_c^5 f(x)\,dx$. We are given that the total area is 12, so $\int_0^5 f(x)\,dx = 12$. We are also given that $\int_0^c f(x)\,dx = k$. Substituting these values.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{(x-6)^n}{n \cdot 4^n}$.

a. ∞

b. $\frac{1}{4}$

c. 6

d. **4** ✓

Applying the ratio test yields $\lim_{n\to\infty} \left|\frac{u_{n+1}}{u_n}\right| = \frac{|x-6|}{4} < 1 \Rightarrow R = 4$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the radius of convergence of the Maclaurin series for $\frac{1}{1-x}$? (variant 3)

a. ∞

b. **1** ✓

c. 2

d. 0

The Maclaurin series for $\frac{1}{1-x}$ has radius of convergence $R = 1$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A differentiable function f is defined on the interval $[-4, 4]$. The graph of f', the derivative of f, is shown in the figure above. The graph of f' consists of two line segments and a semicircle. (a) Find the value of $f(2) - f(-2)$. Show the setup for your answer. (b) For how many values of c in the interval $(-4, 4)$ does $f'(c) = 0$? Explain your reasoning. (c) The function f has a local maximum at $x = k$. Find the value of k. Explain your reasoning. (d) The function f has an inflection point at $x = p$. Find the value of p. Explain your reasoning.

a. There are no values of c for which $f'(c) = 0$.

b. There is one value of c for which $f'(c) = 0$.

c. **There are two values of c for which $f'(c) = 0$.** ✓

d. There are three values of c for which $f'(c) = 0$.

(a) By the Fundamental Theorem of Calculus, $f(2) - f(-2) = \int_{-2}^{2} f'(x)\,dx$. The region from $x = -2$ to $x = 2$ is a semicircle with radius 2. The area of this semicircle is $\frac{1}{2}\pi(2)^2 = 2\pi$. Therefore, $f(2) - f(-2) = 2\pi$. **(b)** The graph of $f'(x)$ is shown. $f'(c) = 0$ when the graph of f' intersects the x-axis. This occurs at $x = -4$ and $x = 2$. Since we are looking for values of c in the open interval $(-4, 4)$, there is one such value, $c = 2$. **(c)** A local maximum o.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.02 Vector-Valued Functions

A particle moves along a straight line so that its velocity at time $t \geq 0$ is modeled by $v(t) = 6t^2 - 18t + 12$. At what time t is the acceleration of the particle equal to 0?

a. $t = \frac{3}{2}$ ✓

b. $t = 1$

c. $t = 2$

d. $t = 3$

To find the acceleration $a(t)$, take the derivative of the velocity function $v(t)$ with respect to time t: $a(t) = v'(t) = \frac{d}{dt}(6t^2 - 18t + 12) = 12t - 18$. Set $a(t) = 0$ and solve for t: $12t - 18 = 0 \Rightarrow 12t = 18 \Rightarrow t = \frac{18}{12} = \frac{3}{2}$. Common Misconceptions: - Solving $v(t) = 0$ yields $6(t-1)(t-2) = 0 \Rightarrow t = 1$ or $t = 2$. These are the times when the particle is at rest, not when its acceleration is zero. - Omitting the factor of 2 when applying the po.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Polar Curves and Derivatives

Consider the function $f(x)$ whose graph is shown above. The graph of $f(x)$ consists of line segments and a quarter circle. Let $g(x) = \int_0^x f(t)\,dt$.

a. **g'(2) = 2** ✓

b. g'(2) = 0

c. g'(5) = -2

d. g'(5) = 0

By the Fundamental Theorem of Calculus Part 1, $g'(x) = f(x)$. We need to find $g'(2)$, which means we need to evaluate $f(2)$. From the graph, at $x = 2$, the function $f(x)$ is on a line segment connecting $(0, 4)$ and $(4, 0)$. The equation of this line segment can be found. The slope is $m = \frac{0-4}{4-0} = -1$. Using the point-slope form with $(4, 0)$, we get $y - 0 = -1(x - 4)$, which simplifies to $y = -x + 4$. Therefore, $f(x) = -x + 4$ for $0 \leq x \leq 4$. Evaluating $f(2)$ gives $f(2) = -2 + 4 = 2$..

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.02 Area in Polar Coordinates

If $f(x) = x^2 + 3$ is increasing on $[1, 6]$, a left Riemann sum with equal subintervals will:

a. equal the true area exactly

b. **underestimate the true area** ✓

c. overestimate the true area

d. cannot be determined

For an increasing function, left-endpoint heights are too small and right-endpoint heights are too large, so a left sum will underestimate the true area.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

If $f(x) = 2x^2 + 5$ is increasing on $[2, 5]$, a right Riemann sum with equal subintervals will:

a. equal the true area exactly

b. cannot be determined

c. underestimate the true area

d. **overestimate the true area ✓**

For an increasing function, left-endpoint heights are too small and right-endpoint heights are too large, so a right sum will overestimate the true area.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Using a right Riemann sum with 4 subintervals of equal width on $[0, 2]$, approximate $\int_0^2 (x^2 + x - 2)\,dx$.

a. $\frac{21}{4}$

b. $-\frac{3}{4}$

c. $-\frac{15}{4}$

d. $\frac{9}{4}$ ✓

$\Delta x = \frac{1}{2}$; heights at the right endpoints give $\frac{1}{2}(-\frac{5}{4} + 0 + \frac{7}{4} + 4) = \frac{9}{4}$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Using the Trapezoidal Rule with 2 subintervals on $[0, 6]$, approximate $\int_0^6 (2x^2 - 3x - 1)\,dx$.

a. 183

b. 21

c. 104

d. **102 ✓**

$T = \frac{\Delta x}{2}[f(x_0) + 2f(x_1) + f(x_2)] = \frac{3}{2}[-1 + 2(8) + 53] = 102$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Integration by Parts

The table shows values of $g(x)$ at $x = [0, 3, 6, 8]$: $g = [2, 3, 8, 7]$. Using a right Riemann sum with these subintervals, approximate $\int_0^8 g(x)\,dx$.

a. 41

b. **47 ✓**

c. 31

d. 50

Sum of (width × right height) over each subinterval: $3(3) + 3(8) + 2(7) = 47$.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

The table shows values of $g(x)$ at $x = [0, 2, 5, 7]$: $g = [7, 8, 12, 4]$. Using a left Riemann sum with these subintervals, approximate $\int_0^7 g(x)\,dx$.

a. 56

b. 65

c. **62** ✓

d. 60

Sum of (width × left height) over each subinterval: $2(7) + 3(8) + 2(12) = 62$.

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